While Loop

```
int main()
{

    //while loop
    //1. initialization of loop variable
    int i=0;
    //2. termination condition
    while(i<5) //0<5 =>1//1<5=>1//2<5=>1//3<5=>1//4<5=>1//5<5=>0
    {
   true     //3. body of loop(code / logic)
        printf("\n Hello Students");
        printf("\n Hello Sunbeam \n");
        //4. modification of loop var(increment/ decrement)
        i++; //i=i+1=> i=0+1=>1+1=>2+1=>3+1=>4+1=>5
    }
false
while has been terminated
}
```

while loop infinity

```
    //1
    int i=0;
    //2
    //infinite loop
    while(i<5); //0<5=>1 //0<5=>1
    {
        //3
        printf("\n Hello Sunbeam");
        //4
        i++;
    }
```

```
    for loop
                        for will end
    int i=1;
    //      11≤10
    for( ; i≤10 ; i++);
    {
        printf("\n %d", 2*i); 2*11=>22
    }
```

do .... whhile ... loop

```
int i=1;
  do
  {
     //2.
     printf("\n %d", 2*i);      2*1⇒2    2*10⇒20
                                2*2⇒4
                                2*3⇒6
     //3.
     i++; //1+1⇒2+1⇒3+1⇒4 . . . 10+1⇒11

  }while(i≤10); //4.
     2≤10 ⇒ 1
     3≤10 ⇒ 1

     10≤10⇒1
     11≤10⇒0
```

true

false

do while loop will be terminated

break stmt

```
int i=0;
  while(i<5)//0<5⇒1
  {
     printf("\n Hello Sunbeam Pune");
     printf("\n Hello Sunbeam Karad");
     break; // while will be terminated
     printf("\n Hello Sunbeam Marketyard"); // will not be executed
     printf("\n Hello Sunbeam Kolhapur\n"); // will not be executed
     i++;
  }
```

while loop will be terminated

continue

```
int i=0;
  while(i<5)//5<5⇒0
  {
     printf("\n Hello Sunbeam Pune");
     printf("\n Hello Sunbeam Karad\n");
     i++; 1 2
     continue; // it will not terminate the loop but it will skip the following stmts
     printf("\n Hello Sunbeam Marketyard"); // will not be executed (skipped)
     printf("\n Hello Sunbeam Kolhapur\n"); // will not be executed (skipped)

  }
```

```
for(int i =1; i≤10; i++)   // 1+1=>2
{
    //3. body of for loop
    printf("\n %d", 2 * i);
}
```

2*1=>2

2*2=>4

terminated

break

```
for(int i=0;i<5; i++)
  {
    printf("\n Hello Students");

    break;// jump stmt break will terminate the for loop

    printf("\n Hello Sunbeam\n"); //this stmt will be skippe

  }
```

for loop will be terminated.

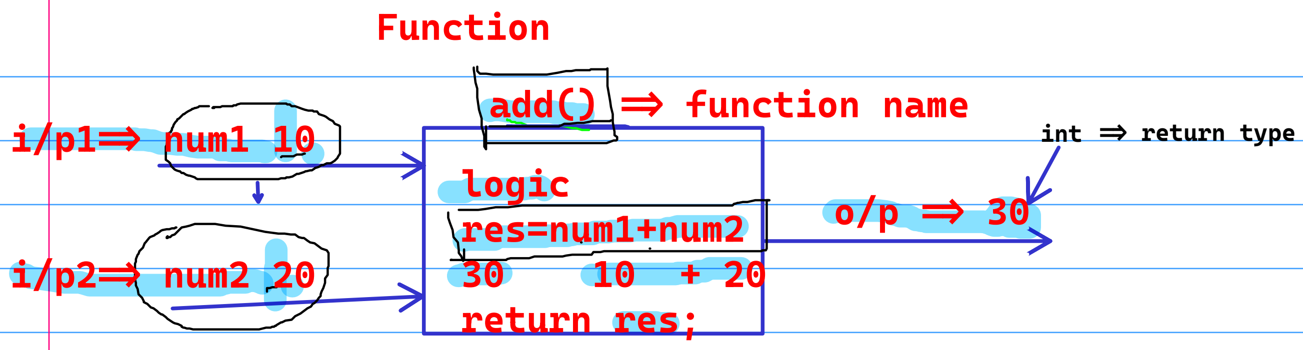
continue

```
for(int i=0;i<5; i++) 0+1 . . . 5
{
  printf("\n Hello Students");

  continue;// jump stmt continue will continue the for loop,
  //skip the stmt after continue

  printf("\n Hello Sunbeam\n"); //this stmt will be skipped(n
}
```

for loop terminated.



1. function declaration/ forward declaration(returnType functionName(formalArguments));
   int add(int num1, int num2);
2. function definition (returnType functionName(formalArguments){logic/code})
   int add(int num1, int num2)
   { int res= num1+num2; return res; }
3. function call (functionName(actualArguments));
   res= add(10,20);

Function

formal argument

```
int add(int  num1, int num2)
{
    //body of add function
    int res= num1+num2;
         //30 + 40 ⇒ 70
         //10 + 20 ⇒ 30
    return res; // result //30//70 will be return
}

int main()
{
    int num1=10, num2=20;
    int res= add(num1, num2); //3. function call(functionName(actualArguments))
                  // 10     20
}
```

local variables

actual argument